

HAYLEY BLOCH

Cambridge, MA | +1 (310) 303-4879 | hayleybloch@college.harvard.edu
[linkedin.com/in/hayleybloch](https://www.linkedin.com/in/hayleybloch) | hayleybloch.com (engineering portfolio)

EDUCATION

Harvard College | Cambridge, MA

Graduating May 2027

A.B. Candidate, History & Science (Engineering Sciences Track)

Related Coursework: How to Make (Almost) Anything (MIT), Computer-Aided Machine Design, Informal Robotics, Innovation in Science and Engineering, Technology Ventures, Intro to Computer Science

RELEVANT EXPERIENCE

Undergraduate Researcher - MIT Media Lab - Center for Bits and Atoms (CBA)

Jan 2026 - Present

- Research rapid-prototyping motion systems, improving the open-source coreXYZ mechanism by reducing moving mass and exploring belt-drive alternatives to cable systems.

Robotics Engineering Intern - NASA Ames Research Center

June 2026 - August 2026

- Designed, prototyped, assembled, and tested robotic construction technologies for lunar infrastructure, integrating robotics, mechanism design, embedded electronics, and additive manufacturing.

Teaching Assistant - Harvard Innovation Labs - Maker Studio

June 2025 - Present

- Oversee training, mentorship, and technical support; assisting students with digital fabrication, prototyping, and safe equipment use while supporting project development across disciplines.

Hardware Engineer, Product and Operations - Halo

April - May 2025

- Led hardware development and operational strategy for an early-stage wearable tech startup using facial recognition; collaborated directly with founders to shape product direction, build prototypes, and scale systems.
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TECHNICAL SKILLS

CAD: SolidWorks, Fusion 360, AutoCAD, Inventor

Fabrication: Laser cutting, CAM, 3D printing, PCB design, embedded systems

Programming: Python, JavaScript, C#

Design: Google UX Professional Design Certification, Blender, Wix, Canva, Figma, Adobe Photoshop, Adobe Illustrator

SELECTED ENGINEERING PROJECTS (PORTFOLIO LINK ABOVE)

Animatronic Humanoid Head

- Designed a 10-servo expressive robotic head with embedded control electronics and wireless interface
- Built custom servo control architecture using ESP32 and PWM driver boards

Digital Wooden Record Player

- Built laser-cut record player using RFID tags mapped to Spotify API
- Designed mechanical enclosure and electronics integration

Punch-Activated Flame Thrower Glove

- Developed IMU-based punch detection system controlling an actuated output via embedded microcontroller

Computer Vision Mechatronic Hand

- Implemented OpenCV-based hand tracking with mirrored servo actuation
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LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Eleganza Show - Executive Producer

Sept 2024 - Present

- Oversee the strategy, branding, and execution of all marketing and communications for Harvard's largest student-run event costing ~\$60,000 and reaching 1,500+ attendees.

Harvard Undergraduate Automotive Society (HUAS) - Executive Board

Sept 2024 - Sept 2025

- Manage HUAS budget and resources, secure FSAE team sponsorships, create and maintain the website and social media, and assist with welding and fabrication of the Formula 1 competition car.